ODSC East AI Accelerator Capstone Project

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Intentions

Goal:
To build a tool to gather the latest Veterinary eCommerce industry information, newly released products, and other external trend data from pre-determined sources. Use AI to gather, summarize and format the content into a structured weekly brief.

Audience:
eCommerce team, with potential for disseminating information to leadership or other teams as appropriate.

Why it Matters:
Staying ahead of Industry trends and identifying risks/opportunities in a timely manner allows us to take actions to better grow our business and support our community.

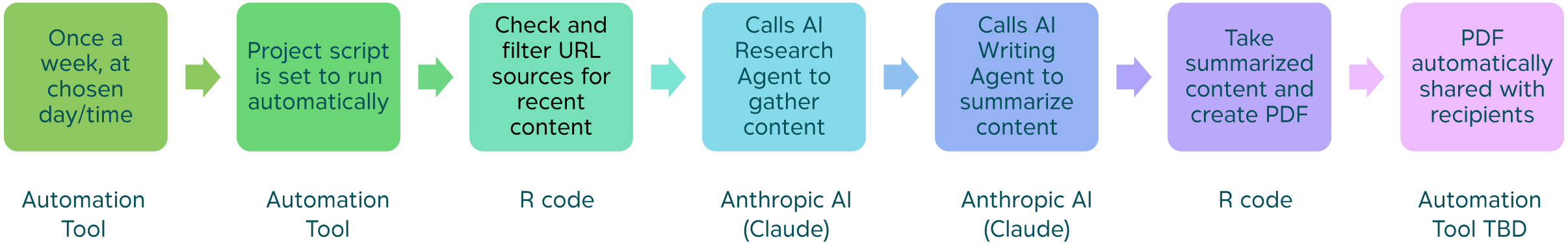
Role of AI in This Project:
Save hours of employee that would otherwise be spent checking web articles for recent content and writing the brief. Being built with tight control prompts in AI agent setup and controlled by a strong R code foundation will ensure consistent results.

Industry Sources Chosen

Well-known & popular sources of Veterinary Industry news and Pet Product articles were chosen as the heart of the brief.

- DVM360
- AVMA.org
- Veterinary Practice News
  - general news and also business/finance page were chosen
- Clinician's Brief
- Cornell University Canine Health Center News
- Pet Insight
- Brakke Consulting
- FDA.gov
- Global Pet
- PetFoodIndustry
- PetFoodProcessing
  - General feed and weekly wrap-up, as well as events, manufacturer news, recalls, corporate news and financial pages were all chosen

Core User Flow Runs on Its Own



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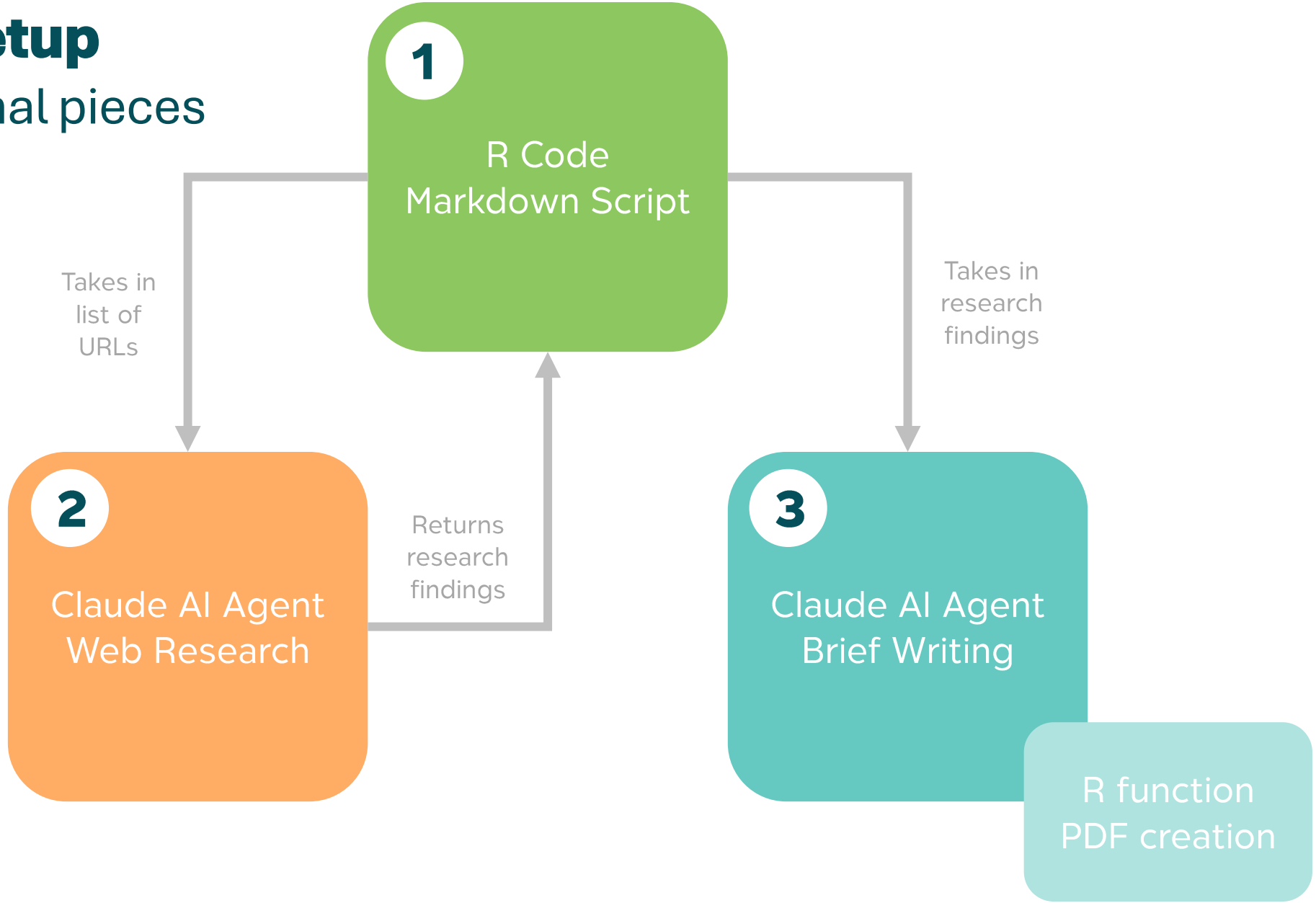
Project Setup

3 foundational pieces



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Workload Breakdown

RMD script

- Determines sources of industry information
- Gets API key, working directory, output directory.
- Sets up Claude model selection(s)
- Filters industry sources down to include only those with available recent content
- Calls Research Agent
- Calls Writing Agent
- Logs token usage and cost for every Agent call

Research Agent

- Takes in a source URL passed from the RMD loop — fetches page content directly using Claude's native web_fetch tool
- Returns structured JSON with article summary, verbatim quotes, publication date, and cited URL per finding.

Writing Agent

- Takes in cleaned, date-filtered, citation-verified research output table from the RMD
- Prompts Claude to compile a Veterinary eCommerce Exec brief from the identified content: transforming it into insights, risks, and opportunities. Adding recommended actions at the end of each section.
- Runs PDF creation function (stored in separate R script)

Agent Settings

Research Agent

Model: Sonnet 4-6
Temperature: 0.1
Claude native tools: web_fetch
Called: once per URL
Cache Prompt?: Yes
Chunking?: No
Return format: JSON
Max Tokens: 3,000
Max Content Tokens: 8,000 tokens per page fetch
Budget Tokens: (not set)
Thinking: (not set)
Timeout: 600 sec
Limits: Summarize in 1-2 sentences,
quotes <= 25 words
Max Uses: 1 fetch per source call

Writing Agent

Model: Sonnet 4-6
Temperature: Omitted: incompatible w/ extended thinking
Claude native tools: (none)
Called: once w/ all content as single input table
Cache Prompt?: No
Chunking?: No (preserve context of entire input)
Return format: JSON -> R function writes PDF
Max Tokens: 16,000
Budget Tokens: 8,000
Thinking: interleaved-thinking-2025-05-14
Timeout: 600 sec
Limits: Sections <= 6 bullets,
Bullets <= 25 words & 1-2 sentences,
Recommended Actions <=3 sentences.

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Balancing RAG vs Context-Grounded Generation

What RAG would look like here: A true RAG implementation would separate the Research Agent into two steps — first retrieving and storing raw web content in a vector database, then retrieving only the most semantically relevant chunks before passing them to a generation step. This adds an embedding model, a vector store, and a retrieval layer

- web_fetch returns page content directly into Claude's context window, where **retrieval and generation happen in a single agent call**, keeping the architecture simple and costs low
- Separating retrieval and generation would require an embedding model, vector store, and retrieval layer - significant added complexity for a weekly brief with a small, structured input table
- **Grounding is enforced through prompt constraints** instead of retrieval filtering. The agent is explicitly instructed to cite sources, return verbatim quotes, and discard content outside the date window
- The R-side freshness filter and quality filter act as a lightweight retrieval layer. **Only verified, recent, cited content reaches the Writing Agent**

A larger source list, context window pressure, or the need to query historical briefs could make a full RAG implementation worth revisiting in the future

Tools & Resources Utilized

Core code

- R language. RMD file for main script, R files for agents and PDF function.

API used for agents themselves

- Anthropic (Claude)

AI chat agents used for vibe coding and error debugging

- Claude - best at vibe coding
- Gemini - another good choice for vibe coding
- Copilot - only option where I have enterprise license available, so messages are unlimited. Used particularly for vibe coding when creating the PDF writing R function and tweaking formatting

Other Resources

- https://skills.sh/ referenced for brief-writing skill guidelines
- Anthropic documentation
- Notes from ODSC Accelerator courses

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Costs & Token Usage

In a practice run on all input URLs:

Research Agent

Input tokens: 1,119

Cache writes: 63,754

Cache reads: 196,119

Total input: 260,992

Output tokens: 3,189

Cache hit rate: 75.1 %

Cost?: $0.12

Writing Agent

Input tokens: 1,052

Output tokens: 3,269

Cost?: $0.02

Feature	Claude Opus 4.7	Claude Sonnet 4.6	Claude Haiku 4.5
Description	Our most capable generally available model for complex reasoning and agentic coding	The best combination of speed and intelligence	The fastest model with near-frontier intelligence
Pricing[1]	$5 / input MTok $25 / output MTok	$3 / input MTok $15 / output MTok	$1 / input MTok $5 / output MTok
Extended thinking	No	Yes	Yes
Adaptive thinking	Yes	Yes	No
Context window	1M tokens	1M tokens	200k tokens
Max output	128k tokens	64k tokens	64k tokens
Reliable knowledge cutoff	Jan 2026[2]	Aug 2025[2]	Feb 2025
Training data cutoff	Jan 2026	Jan 2026	Jul 2025

1 - See the pricing page for complete pricing information including batch API discounts, prompt caching rates, extended thinking costs, and vision processing fees.

*2 - **Reliable knowledge cutoff** indicates the date through which a model's knowledge is most extensive and reliable. **Training data cutoff** is the broader date range of training data used. For more information, see Anthropic's Transparency Hub.*

3 - Claude Opus 4.7 on AWS is available through Claude in Amazon Bedrock (the Messages-API Bedrock endpoint).

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